A Refined Estimator for Fraud Detection Under Distribution Shift

Abstract

This report summarizes a study on making *conformal prediction* (CP) more reliable for fraud-detection systems whose data distribution drifts over time. The central idea is to pair a legacy classifier with a small, freshly retrained one and to learn an auxiliary *quality model* that estimates where each classifier is likely to be wrong; the two predictors are then blended into a single *refined* score that is fed into an off-the-shelf split-conformal routine. On synthetic data the refined estimator improves coverage on the hardest (most drifted) transactions while keeping prediction sets small on easy ones, and a real transaction dataset shows the same qualitative behaviour under label shift. The guarantees are empirical rather than theoretical, several design choices were fixed by hand rather than tuned.

1 Motivation

Conformal prediction [1, 2] is a distribution-free wrapper that turns any point predictor into one that outputs *prediction sets* with finite-sample marginal coverage. Because it needs only a model's scores on a held-out calibration set, it works with any black-box classifier.

Standard split-conformal, however, keeps the base model's score ordering and merely picks a single global threshold. When the classifier is *over-confident* — assigning very high probability to wrong classes — CP can only satisfy coverage by inflating *all* sets or by still missing the truth on confident mistakes [3, 4].

This is exactly the regime of fraud detection. Purchasing behaviour and attack patterns drift over time (concept / dataset shift [5]), and when performance degrades operators retrain on the small batch of recently labelled transactions [6]. That sparse supervision often leaves the updated model over-confident, so plain CP either overloads analysts with large sets or misses the rare, high-confidence errors. We ask whether a light-weight adjustment built from the data already available can do better.

2 Method

Ingredients.

- $\hat{\pi}_{\text{old}}$: legacy classifier trained once on a large sample $S_{\text{old}} \sim P_{\text{old}}$.
- S_{new}: a *small* fresh sample from P_{new}, split into train / calibration / test.
- $\hat{\pi}_{\text{new}}$: a quick retrain on the train split of S_{new}.
- $\hat{\gamma}_{\text{old}}$: a *quality model* estimating the probability that $\hat{\pi}_{\text{old}}$ is wrong at a given input.

Quality model. The quality model is trained on a labelled set $\{(X_i, Y_i)\}_{i=1}^n$ and a base classifier $\hat{\pi}$ by the following recipe:

> **Algorithm (quality-model training).** For each i: compute $\hat{p}_i = \hat{\pi}(X_i)$; draw a pseudo-label $\tilde{Y}_i \sim \text{Bernoulli}(\hat{p}_i)$; set the mismatch indicator $M_i = \mathbf{1}\{\tilde{Y}_i \neq Y_i\}$. Then fit a binary learner predicting M_i from X_i and return $\hat{\gamma}(x) = \Pr(M = 1 \mid X = x)$.

Thus $\hat{\gamma}_{\text{old}}(x)$ is an estimate of the legacy model's error probability at x. For simplicity we *do not* learn a separate quality model for the new classifier and instead fix $\hat{\gamma}_{\text{new}}(x) = \frac{1}{2}$, a neutral value.

Refined estimator. We first temper the new model toward $\frac{1}{2}$ (curbing its over-confidence),

$$\hat{\pi}'_{\text{new}}(x) = \big(1 - \hat{\gamma}_{\text{new}}(x)\big)\, \hat{\pi}_{\text{new}}(x) + \hat{\gamma}_{\text{new}}(x) \cdot \tfrac{1}{2} \xrightarrow{\hat{\gamma}_{\text{new}} = \frac{1}{2}} \tfrac{1}{2}\, \hat{\pi}_{\text{new}}(x) + \tfrac{1}{4}, \tag{1}$$

and then blend old and new, gated by the legacy quality model:

$$\boxed{\hat{\pi}_{\text{refined}}(x) = \big(1 - \hat{\gamma}_{\text{old}}(x)\big)\, \hat{\pi}_{\text{old}}(x) + \hat{\gamma}_{\text{old}}(x)\, \hat{\pi}'_{\text{new}}(x)}. \tag{2}$$

Intuitively: when $\hat{\gamma}_{\text{old}}(x)$ is low the seasoned legacy model is trusted; when it is high the fresh model is allowed to contribute, but only after its over-confidence has been scrubbed away. Because the weights are bounded and data-dependent, the blend stays well-behaved even if $\hat{\pi}_{\text{new}}$ is poorly calibrated. The refined score is then passed unchanged into standard split-conformal calibration, so the usual finite-sample coverage machinery still applies.

3 Synthetic experiments

Data-generating process. We simulate transactions with three components (three Gaussians with a shared covariance), which differ in how fraudulent they are:

$$P^{(0)} : \text{Ber}(0) \text{ (safe)}, \qquad P^{(1)} : \text{Ber}(1) \text{ (known fraud)}, \qquad P^{(2)} : \text{Ber}(\rho_2),\ 0 < \rho_2 < 1 \text{ (new behaviour)}.$$

The old world mixes the clean components and the new world introduces the ambiguous one:

$$P_{\text{old}} = c_{\text{old}} P^{(0)} + (1 - c_{\text{old}}) P^{(1)}, \qquad P_{\text{new}} = c_{\text{new}} P^{(0)} + (1 - c_{\text{new}}) P^{(2)}.$$

We label test points drawn from $P^{(2)}$ as **hard**: they represent patterns no model has truly mastered.

Set-up. We draw 10,000 legacy and 5,000 fresh samples with $\rho_2 = 0.6$, $c_{\text{old}} = 0.8$, $c_{\text{new}} = 0.7$, split 60/20/20 (train / calibration / test). Every number below is a mean $\pm$ s.e. over 100 Monte-Carlo repeats; the target coverage is $1 - \alpha = 0.9$.

Table 1: Conformal prediction sets on synthetic fraud data, stratified by test difficulty. $\overline{|C|}$ is the average set size (1 = single-label prediction); Cov. is empirical coverage (target 0.9). **Refined** is the proposed estimator.

	Easy		Full		Hard	
Model	$\overline{\lvert C\rvert}$	Cov.	$\overline{\lvert C\rvert}$	Cov.	$\overline{\lvert C\rvert}$	Cov.
π_{old}	1.24	1.000	1.30	0.900	1.43	0.669
π_{new}	1.08	0.993	1.22	0.938	1.54	0.812
Refined	**1.03**	0.998	1.22	0.953	1.66	**0.848**

Two things stand out. On the *hard* split the legacy model badly undercovers (0.669), the retrained model does better (0.812), and the refined estimator is closest to target (0.848) — buying that coverage with modestly larger hard sets. On the *easy* split it also produces the smallest sets (1.03) while staying above target, i.e. it spends set size where it is actually needed.

Sweeping the hidden fraud rate. Figure 1 sweeps $\rho_2 \in \{0.4, \ldots, 1.0\}$ at $n_{\text{new}} = 5{,}000$. As ρ_2 grows the legacy model's sets widen (it was calibrated on a lower fraud prevalence), whereas the refined combiner keeps coverage near 0.9 across the range while producing sets comparable to or smaller than the legacy model.

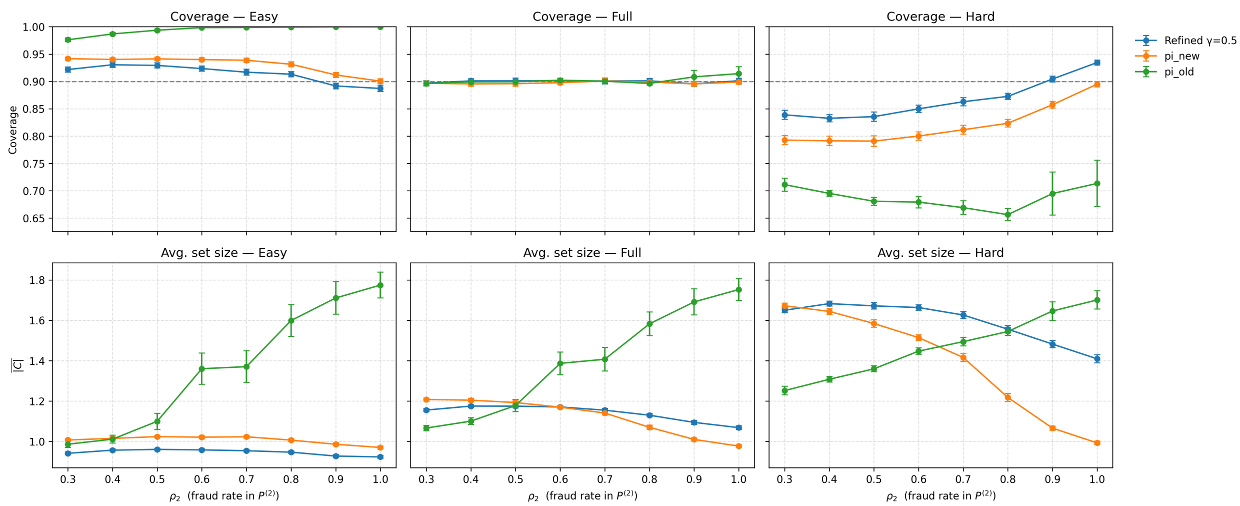

Figure 1: Split-conformal *coverage* (top row) and average set size $\overline{|C|}$ (bottom row) versus ρ_2, on the easy / full / hard splits, for the legacy model (π_{old}), the retrain (π_{new}), and the refined combiner ($\hat{\gamma}_{\text{new}} = \frac{1}{2}$). Fresh sample size $n_{\text{new}} = 5{,}000$; error bars are $\pm 1.96\,\text{SE}$ over 30 repeats; mixture weights $c_{\text{old}} = 0.8$, $c_{\text{new}} = 0.7$.

Comparison against baselines. Figure 2 compares the refined predictor (`Quality_Cond`) against standard CP on each model and against trust-score conditional coverage [3] (`Trust_Cond_*`), as the new-data size grows. On the hard subset our method tracks the nominal level most closely while keeping set sizes competitive; all methods stabilize as $|S_{\text{new}}|$ increases.

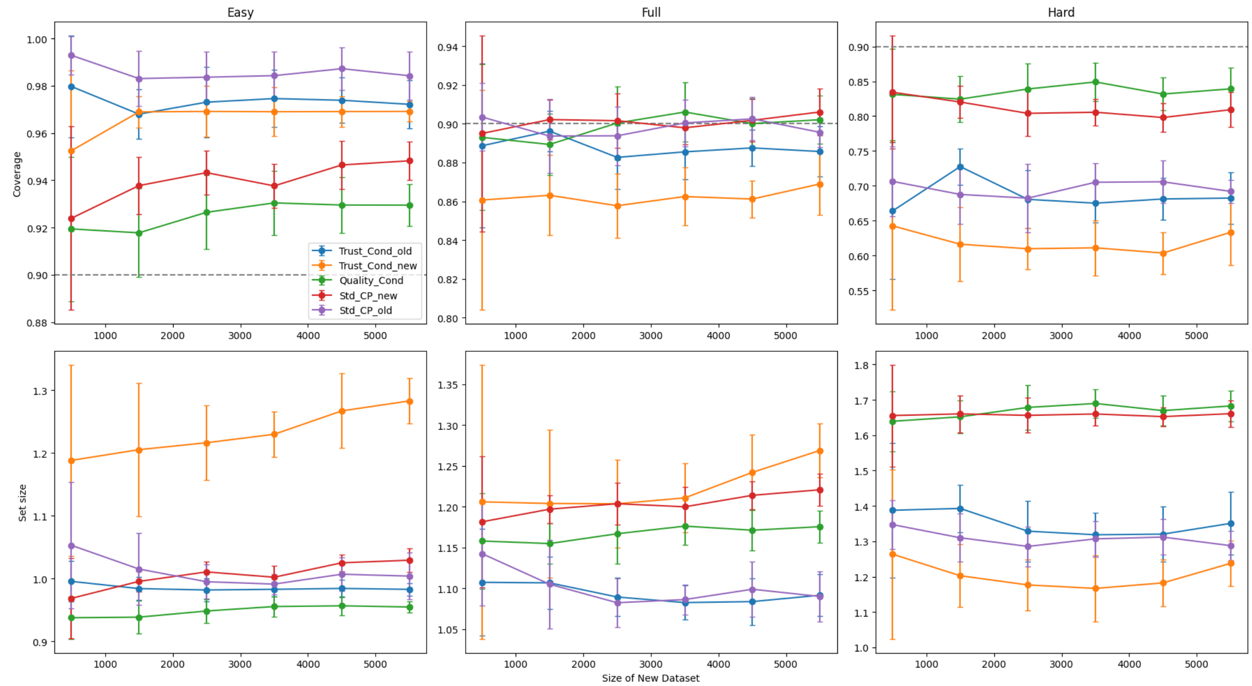

Figure 2: Coverage (top) and average set size (bottom) on Easy / Full / Hard subsets as a function of the new-dataset size, at target coverage 0.9 (dashed). `Quality_Cond` is the proposed refined predictor; `Trust_Cond` and `Std_CP` are the trust-score and standard-CP baselines on the old and new models. Averaged over 10 seeds.

4 Real-data experiments

We also ran a smaller study on a real transaction dataset (features such as transaction amount, origin/destination balances, a time-of-day encoding, and transaction type; each sample labelled fraud / non-fraud). To emulate drift we split the data into an **OLD** and a **NEW** distribution, trained a model on each, and evaluated `Std_CP_old`, `Std_CP_new`, and the refined `Refined_CP` on Easy / Full / Hard partitions of the NEW data.

Covariate shift. Here the feature distribution changes but $P(Y \mid X)$ is held fixed. On this dataset the covariate shift turns out to be mild (Figure 3): once enough NEW samples are available all three methods reach similar coverage, and the refined predictor is mainly useful as a stabilizer at small sample sizes. This is an honestly negative-ish result — the OLD and NEW distributions here are fairly aligned.

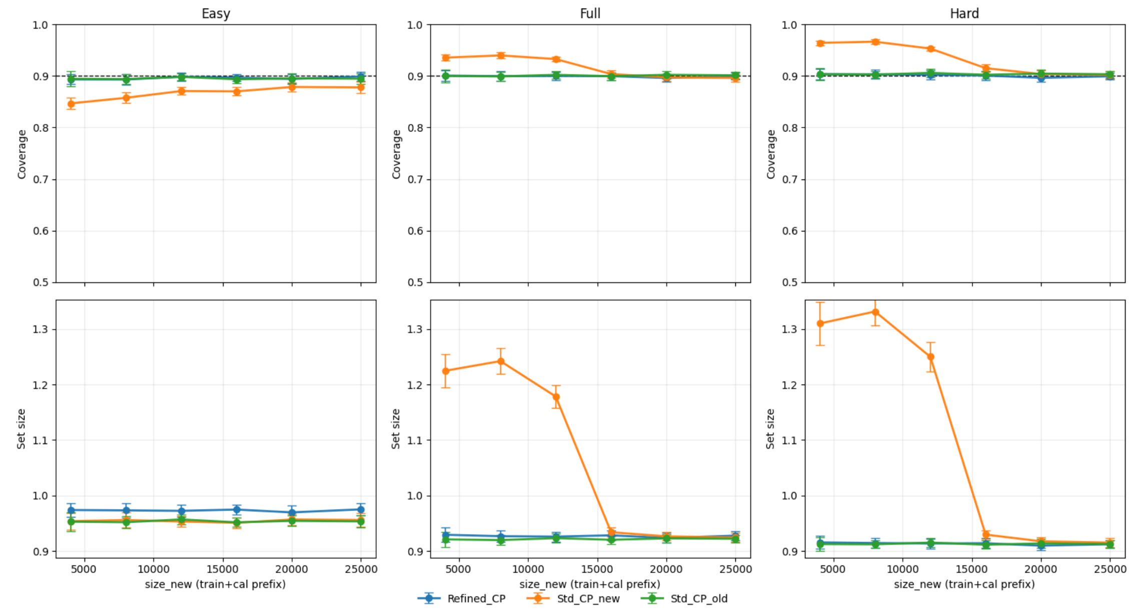

Figure 3: Covariate-shift experiment on real data: coverage (top) and average set size (bottom) across Easy / Full / Hard subsets. The shift is weak, so the methods behave similarly once calibration data is sufficient.

Label shift. When we instead change the label distribution in selected regions (simulating newly emerging fraud patterns), the OLD model's confidence drops sharply on hard samples whose features previously looked non-fraudulent. The refined predictor, by adaptively blending OLD and NEW, achieves better calibration and smaller coverage gaps in exactly those hard regions (Figure 4).

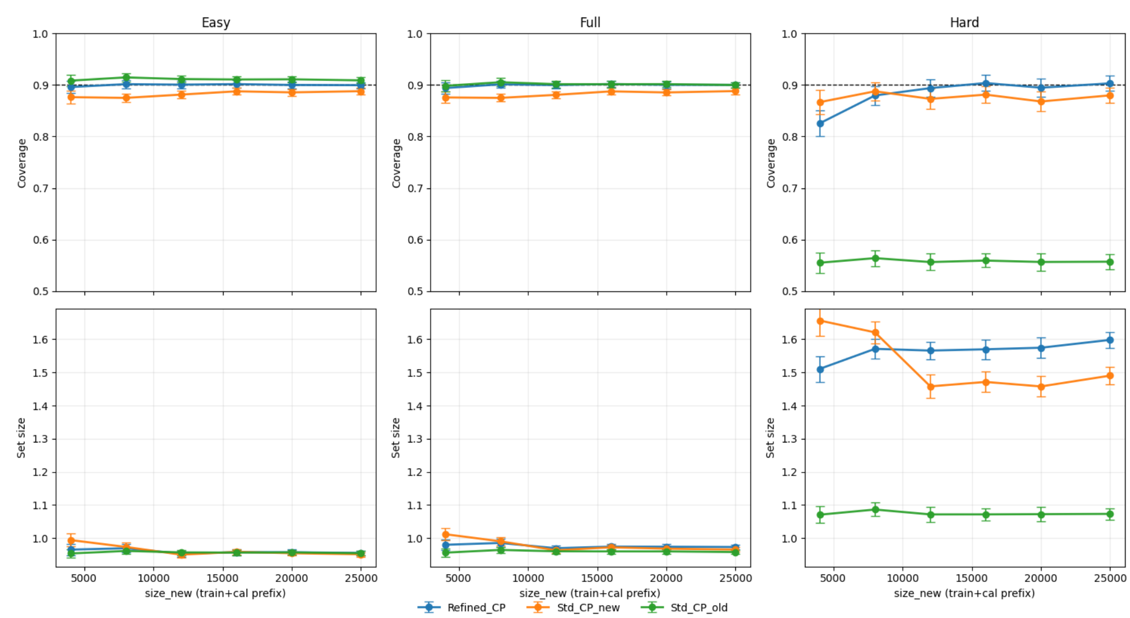

Figure 4: Label-shift experiment on real data: the refined predictor improves conditional coverage on the hard subset relative to standard CP on either single model.

5 Limitations

This is preliminary work and several pieces are deliberately simple:

- **No theory yet.** The coverage improvements are empirical. Because the refined score is fed into ordinary split-conformal, *marginal* coverage holds by construction, but the *conditional* (per-difficulty) behaviour is only observed, not proven.
- **Hand-fixed choices.** The new-model temperature is fixed at $\hat{\gamma}_{\text{new}} = \frac{1}{2}$ and the tempering target is $\frac{1}{2}$; these were not tuned.

Overall, the results suggest that a learned quality model plus a bounded old/new blend is a cheap and promising way to protect conformal fraud detectors where they fail most — on confident, drifted transactions.

References

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